Naitik Patel

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.41
SHREE VASISTHA VIDHYALAYA	Class XII (GSEB)	2023	80.60%
SHREE VASISTHA VIDHYALAYA	Class X (GSEB)	2021	96.67%

SKILLS

Programming Languages: C, C++ (Proficient); Python, Java, SQL (Familiar)

Problem Solving: Proficient in Data Structures and Algorithms with hands-on problem-solving experience.

Core CS Fundamentals: Operating Systems, Database Management Systems (DBMS)

ML/DL : Supervised and Unsupervised, Transformers, Foundational Knowledge of LangChain and LLMs.

Integrated Development Environments: Visual Studio Code, Jupyter Notebook, Google Colab

Libraries: Numpy, Pandas, Matplotlib, TensorFlow, Sklearn

Tools: Git, GitHub, SolidEdge

Soft Skills: Teamwork, Adaptability, Leadership, Analytical Thinking, Communication, Management

PROJECTS

AI-Powered Document Q&A System (RAG Chatbot) -- Python, LangChain, FAISS, Gemini, Groq (GitHub)

- Engineered an end-to-end **RAG chatbot** supporting **3 file formats** (PDF, TXT, CSV) by generating vector embeddings via **Google Gemini** and indexing them in **FAISS** for semantic search
- Integrated **Groq's LLaMA 3.1** with a strict grounding prompt so that **100%** of answers were sourced exclusively from uploaded documents, eliminating hallucinated responses
- Built a **Streamlit** interface with drag-and-drop uploads, real-time indexing, and persistent chat history with per-answer source citations
- Architected the codebase into **4 modular components** with API rate-limit retry logic across Gemini and Groq free-tier services to ensure reliable uptime

Movie Recommender System -- Python, Streamlit, NumPy, Pandas (GitHub)

- Developed a **content-based recommendation engine** in Python and Streamlit, generating personalized movie suggestions from large-scale metadata
- Implemented **vectorization and cosine similarity** algorithms on movie metadata to power real-time recommendations
- Applied **NLP-based feature engineering** (string concatenation of movie attributes) to improve recommendation accuracy
- Integrated the **TMDB API** to display movie posters within an interactive, real-time web interface

B+Tree-Based Database Storage Engine -- C++17 (GitHub)

- Designed and implemented a **disk-based B+Tree indexing engine** in C++17, achieving **O(log n)** time complexity for insert, search, delete, and range-scan operations
- Built a custom page management system for fixed-size **4KB** disk I/O, ensuring **zero data loss** across application restarts
- Engineered node-splitting and tree-balancing logic -- including recursive internal-node splits and new-root creation -- to support dynamic dataset growth
- Developed a unit test suite of **6+ test cases** covering insertion, deletion, range queries, and persistence; automated

build verification via **CI/CD (GitHub Actions)**

Dynamic Pathfinding System with Multi-Constraint Optimization -- C++, Graph Algorithms (GitHub)

- Designed a tactical **A*-based pathfinding simulator** with custom cost functions for danger avoidance and cover utilization
- Implemented graph algorithms and real-time threat assessment to enable adaptive route optimization
- Integrated multi-constraint balancing logic to select the safest, most efficient path across competing objectives
- Built ASCII-based visualizations to demonstrate algorithmic reasoning and benchmark performance outcomes

ACHIVEMENTS

Competitive Programming: Achieved **1600+ LeetCode Contest Rating** and solved **600+ DSA problems** across LeetCode, GeeksforGeeks, Codeforces, and Coding Ninjas.

RELEVANT COURSES

Computer Science: Introduction to Programming, Introduction to AI/ML.

Mathematics: Calculus, Elementary Linear Algebra, Differential Equations, Numerical Methods, Applied Mathematica in Chemical Engineering(includes various topics on statistics and probability).

POSITIONS OF RESPONSIBILITY

Operations Coordinator | Finance & Consulting Club, IIT Hyderabad|(2024-2025)

- Coordinated planning and execution of finance and consulting events, managing logistics to ensure smooth event operations.
- Organized finance-focused workshops, speaker sessions, and networking events to facilitate student engagement.

Internship Coordinator | Office of Career Services, IIT Hyderabad|(2025-2026)

- Coordinated with recruiters and companies during campus internship recruitment drives.
- Facilitated communication between students and recruiters to support scheduling, queries, and recruitment-related coordination.

EXTRACURRICULAR

Member of **National Service Scheme (NSS)** , IIT Hyderabad.

Actively engage in cricket and chess as personal hobbies.